David Barreiro Salgueiro

Senior Software Engineer | Distributed Systems · Data Platforms · Streaming

A Coruña, ES

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David Barreiro (https://www.linkedin.com/in/david-barreiro-s/)

Personal projects (https://github.com/davicafu)

Senior Software Engineer with hands-on experience designing and building scalable platforms, distributed systems and real-time data solutions. My background combines backend engineering, data infrastructure and cloud-native platforms. I have worked with Java, Go, Kafka, Spark(Python), ClickHouse, Kubernetes, Docker and modern CI/CD environments, delivering reliable systems for demanding workloads. Recently focused on Golang for high-performance backend services and modern platform development. I enjoy simplifying complex problems, improving performance and building practical solutions that scale. Core areas include backend engineering with Go and Java, distributed systems, Kafka-based streaming architectures, cloud-native platforms, APIs, microservices and DevOps automation.

SKILLS

Programming Languages
Advanced

Go |Java |Bash |Python |Scala |
Spring Boot

Databases
Advanced

PostgreSQL |ClickHouse |MariaDb |
Redis |MongoDB |Snowflake

Data Processing
Advanced

Kafka |Flink |Databricks |Spark |
RabbitMQ |DBT |Airflow

AI Engineering
Intermediate

OpenSpec |AI coding workflows |
Agentic Systems |Tool orchestration |
Model Interoperability |
Context Engineering |
Workflow Automation |MCP

Infra & DevOps
Advanced

CI/CD (GitHub, Github Actions, Jenkins, GitLab, EAWS |Azure |Google Cloud
Docker |Kubernetes |Terraform

Cloud
Intermediate

Observability & Analytics
Advanced

Grafana |OpenTelemetry |Prometheus |
Superset |Elastic (ELK) |Power BI

WORK EXPERIENCE

Apr 2022 - Current

Senior Software Engineer at SDG Group

Designed and developed scalable real-time data platforms using Kafka, Spark and containerized environments.

- Creation and maintenance of development images with Docker.
- Message pipeline with Spark for data processing using Scala and Python.
- Development of low-code solution for data exploration with KSQL.
- Built backend services and internal tooling in Go and Java to improve operational efficiency.
- Implementation of Kafka Connect, Debezium, DBT & Airflow for data processing.
- Enhanced observability and reliability through Grafana dashboards, monitoring and automated deployments.
- Contributed to architecture decisions, scalability and maintainability of critical systems.

Mar 2019 - Apr 2022

Software Engineer at Altia

Delivered backend and data engineering solutions for enterprise analytics environments.

- Developed microservices and APIs with Java / Spring Boot.
- Implemented near real-time pipelines with Kafka Streams.
- Built monitoring and observability systems with Prometheus.
- Automated CI/CD and deployment workflows.
- Improved reliability, scalability and operational efficiency.

Jun 2018 - Mar 2019

Software Engineer at Emetel

Built and operated a real-time data processing platform using Kafka, Spark and Elastic Stack.

- Created and maintained Docker images for reproducible deployments.
- Developed Java and Scala services for stream processing.
- Implemented ingestion pipelines from multiple external sources.
- Built Kibana dashboards for monitoring and analytics.
- Automated deployment and operational processes.

Dec 2014 - Jun 2018

Junior Software Engineer at Indra Software Labs S.L.

Managed and developed distributed database platforms for a multinational retail environment

- Administered medium-sized clusters (100–200 servers) supporting large-scale data processing workloads.
- Built internal tools for orchestration and platform operations.
- Implemented ingestion processes from DB2, AS/400 and SQL Server systems.
- Optimized data processing workflows based on MapReduce paradigms.
- Improved platform stability, performance and operational efficiency.
- Collaboration in the management of a SLURM cluster (SimpleLinuxUtiliyforResource Management)

PROJECTS

go-auth-kit

Reusable Go authentication module designed for production-ready backends. It supports configurable authentication strategies (cookie-based sessions or JWT access + refresh tokens), role and permission checks (RBAC), OAuth login providers (GitHub / Google), password reset flows, rate limiting at auth endpoints, and security-oriented audit logs. The objective is to provide a drop-in foundation for secure auth without reimplementing the same patterns in every service.

- Go
- JWT + Cookies
- OAuth
- RBAC
- Audit Logs

Concurrent Worker Runtime

Concurrent processing runtime for asynchronous jobs in Go. It includes dynamic worker pools, exponential retries with jitter, backpressure management, dead-letter queue (DLQ), graceful shutdown, and context cancellation propagation. The focus is reliability under load and predictable behavior during failure scenarios, with clear observability hooks for monitoring throughput, retries, and queue health.

- Concurrency
- Retries
- Backpressure
- DLQ
- Graceful Shutdown

Go Microservice Suite (gRPC + REST)

Reference microservice system composed of User Service, Notification Service, and a mock Billing Service. Internal communication uses gRPC contracts with protobuf, while an external REST gateway exposes public endpoints. It includes timeout and retry policies, transport-level error handling, and service boundaries oriented to real-world distributed systems. Built to demonstrate practical service decomposition and robust inter-service communication.

- gRPC
- REST Gateway
- Protobuf

Event Analytics Pipeline

Event-driven analytics architecture covering event ingestion, data modeling, ETL transformations, metric generation, and analytical serving layers. Implemented with Go and analytical/storage combinations such as DuckDB, ClickHouse, and PostgreSQL. The goal is to provide a pragmatic blueprint for near real-time analytics with clear separation between ingestion, processing, and consumption concerns.

- Event-Driven
- ETL
- Go
- ClickHouse
- PostgreSQL

observekit

Observability demo stack for modern backend systems. It integrates OpenTelemetry instrumentation, distributed tracing, Prometheus metrics, Grafana dashboards, centralized logs, SLO-oriented views, and Superset analytics boards for exploratory analysis. The project is focused on making system behavior observable end-to-end: from request-level traces to service KPIs, alerting, and product-level analytical insights.

- OpenTelemetry
- Tracing
- Prometheus
- Grafana
- Superset

CLI Suite

Professional Go CLI toolkit designed as a multi-command platform for operational and engineering workflows. Planned commands include Data Quality CLI, API Audit CLI, Benchmark CLI, Migration CLI, and Kubernetes helper utilities. The suite emphasizes robust UX, strong error handling, testability, and repeatable automation for day-to-day engineering tasks.

- Go
- Automation
- Benchmarking
- Migrations
- Testing

CLI Games Collection

Progressive terminal game collection used as a technical playground for architecture and algorithmic practice. The roadmap advances from grid-based games (Snake, Sokoban) to physics and action loops (Flappy, Breakout, Space Invaders), and then to more complex systems (Pac-Man style AI, Asteroids movement, platformers, Tetris). It is intentionally structured to improve game loops, input handling, collision systems, and reusable engine patterns.

- Game Loops
- Collision Systems
- Terminal UI
- Algorithms
- Architecture

EDUCATION

2008 - 2014

Degree Computer Engineering at Santiago de Compostela University (USC)

2016 - 2018

Master Big Data and Visual Analytics at International University of La Rioja (UNIR)

CERTIFICATES

2025-09-13

ClickHouse Certified Developer

ClickHouse

https://www.credly.com/badges/09a51399-9a62-4619-9d6b-3096e2b8070c/linked_in?t=t2irdp

2016-10

Oracle Database 11g: Performance Tuning

2017-06

Cloudera: Developer Training for Spark & Hadoop

2020-06

AWS Certified SysOps Administrator

LANGUAGES

Spanish
Native speaker

Galician
Native speaker

English
Intermediate speaker